

Information and Network Security

Study Guide

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2.1 Symmetric Cipher Model

- Terminologies:

plaintext - original message

ciphertext - coded message

Cipher - Encryption algorithm for transforming plaintext to ciphertext

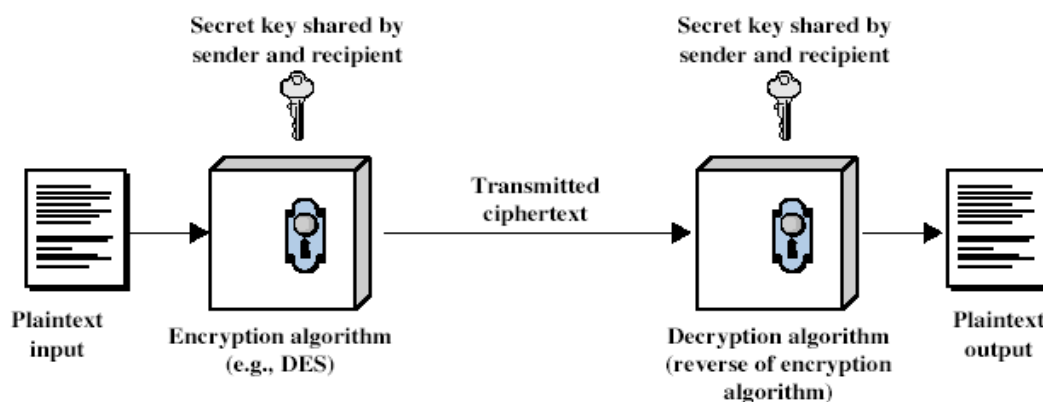
Secret key - info used in cipher known only to sender/receiver

encipher (encrypt)

converting plaintext to ciphertext

decipher (decrypt)

recovering ciphertext from plaintext



Two requirements for secure use of conventional encryption

1. we need a strong encryption algorithm. At a minimum, we would like the algorithm to be such that an opponent who knows the algorithm and has access to one or more ciphertexts would be unable to decipher the ciphertext or figure out the key.

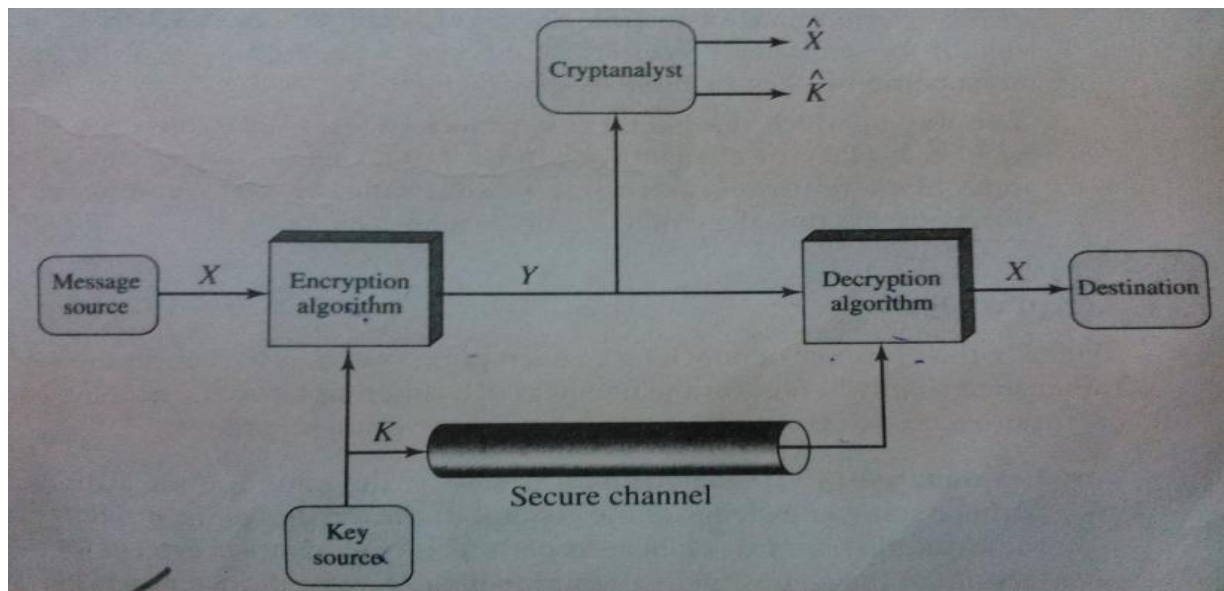
This requirement is usually stated in a strong form: The opponent should be unable to decrypt ciphertext or discover the key even if he or she is in possession of a number of ciphertexts together with the plaintext that produced each ciphertext.

2. Sender and receiver must have obtained copies of the secret key in a secure fashion and must keep the key secure. If someone can discover the key and knows the algorithm, all communication using this key is readable.

- Plain text $X=[x_1, x_2, x_3, \dots]$ Key $K=[k_1, k_2, \dots]$ Cipher text $Y=[y_1, y_2, \dots]$

$$Y=E(K, X)$$

$$X=D(K, Y)$$



References:

1. Cryptography and Network Security (Textbook) William Stallings; Pearson Education
2. Cryptography & Network Security Behrouz A. Forouzan; Tata McGraw-Hill
3. Information Security Principles and Practice Deven Shah, Wiley-India
4. Information Security Principles and Practice Mark Stamp; Wiley India edition
5. Information systems security Nina Godbole; Wiley Publications, 200

2.2 Cryptography

Cryptography is the study of mathematical techniques related to aspects of information security such as confidentiality, data integrity, and entity/data origin authentication.

Characterized in three dimensions:

Type of operation used

- 1) Substitution : Elements of plaintext are mapped into another element
- 2) Transposition : Elements of plaintext are rearranged

The Number of keys used

- 1) Symmetric, Single-key, Secret-key, Conventional Encryption : Same Key is Used
- 2) Asymmetric, two-key, Public-key : different keys are used

Way in which the plaintext is processed

- 1) Stream Cipher : processes the input elements continuously, producing output one element at a time.
- 2) Block Cipher : processes the input one block of elements at a time, producing an output block for each input block.

CRYPTANALYSIS:

- **Cryptanalysis** is the study of mathematical techniques for attempting to defeat cryptographic techniques that provide information security services.
- Cryptanalysis means “**breaking the code**”
- Techniques used for deciphering a message without any knowledge of enciphering details fall into the area of cryptanalysis
- Typically, the objective of attacking an encryption system is to **recover the key** in use rather than simply to **recover the plain text**.
- Two approaches:
 - cryptanalytic attack
 - brute-force attack
- **Brute-force attack**
- A brute-force attack involve trying every possible key until an intelligible translation of the cipher text into plaintext is obtained.

Key Size (bits)	Number of Alternative Keys	Time required at 1 decryption/ μ s	Time required at 10^6 decryptions/ μ s
32	$2^{32} = 4.3 \times 10^9$	$2^{31} \mu s = 35.8$ minutes	2.15 milliseconds
56	$2^{56} = 7.2 \times 10^{16}$	$2^{55} \mu s = 1142$ years	10.01 hours
128	$2^{128} = 3.4 \times 10^{38}$	$2^{127} \mu s = 5.4 \times 10^{24}$ years	5.4×10^{18} years

168	$2^{168} = 3.7 \times 10^{50}$	$2^{167} \mu s = 5.9 \times 10^{36}$ years	5.9×10^{30} years
26 characters (permutation)	$26! = 4 \times 10^{26}$	$2 \times 10^{26} \mu s = 6.4 \times 10^{12}$ years	6.4×10^6 years

- **Cryptanalysis attacks**
- Cryptanalysis attacks rely on
 - the nature of the algorithm plus
 - some knowledge of the general characteristics of the plain text or even some sample plaintext-ciphertext pairs.
- This type of attack exploits the characteristics of the algorithm to attempt to **deduce a specific plaintext** or to **deduce the key** being used.

Types of Cryptanalysis Attacks:

Type of Attack	Known to Cryptanalyst
Ciphertext Only	• Encryption algorithm • Ciphertext
Known Plaintext	• Encryption algorithm • Ciphertext • One or more plaintext-ciphertext pairs formed with the secret key
Chosen Plaintext	• Encryption algorithm • Ciphertext • Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key
Chosen Ciphertext	• Encryption algorithm • Ciphertext • Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key
Chosen Text	• Encryption algorithm • Ciphertext • Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key • Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key

- **Ciphertext only attack**
 - Opponent must rely on an analysis of ciphertext by applying various statistical tests on it
 - Easiest to defend because the opponent has the least amount of information
- **Known plaintext attack (Probable-word attack)**
 - Analyst may be able to deduce the key on the basis of the way in which the known plaintext is transformed.
 - Ex. are
 - there may be a standardized header or banner to an electronic funds transfer message
 - A file that is encoded in the Postscript format always begins with the same pattern
 - If an entire accounting file is being transmitted, the opponent may know the placement

of certain key words in the header of the file

- Organization might include Copyright statement in some standardized position

- **Chosen plaintext attack**

- If analyst is able somehow to get the source system to insert into the system a message chosen by the analyst, then chosen plaintext attack is possible
- If analyst is able to choose the message to encrypt, the analyst may deliberately pick patterns that can be expected to reveal the structure of the key

Unconditional security:

- An encryption scheme is **unconditionally secure** if the cipher text generated by the scheme does not contain enough information to determine uniquely the corresponding plaintext, no matter how much cipher text is available.
- That is, no matter how much computer power or time is available, the cipher cannot be broken since required information is not there.
- There is no encryption algorithm that is unconditionally secure except one-time pad. So,
 - The cost of breaking cipher exceeds the value of encrypted information
 - The time required to break cipher exceeds the useful lifetime of the information

Computational security:

- An encryption scheme is said to be **computationally secure** if either of the following two criteria are met.
- The cost of breaking the cipher exceeds the values of the encrypted information.
- The time required to break the cipher exceeds the useful lifetime of the information.

Symmetric key cryptography:

- **Symmetric encryption** is a form of cryptosystem in which encryption and decryption are performed using the same key. It is also known as **conventional encryption**.
- Symmetric encryption transform plaintext into ciphertext using a secret key and an encryption algorithm.
- Using the same key and a decryption algorithm, the plaintext is recovered from the ciphertext.

References:

1. Cryptography and Network Security (Textbook) William Stallings; Pearson Education
2. Cryptography & Network Security Behrouz A. Forouzan; Tata McGraw-Hill
3. Information Security Principles and Practice Deven Shah, Wiley-India
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2.3 Substitution Techniques

In this the letters of plaintext are replaced by other letters or by numbers or symbols.

Some of the Substitution Techniques are

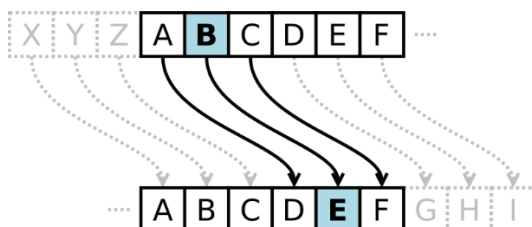
- Caesar cipher
- Monoalphabetic Cipher
- Playfair cipher
- Hill cipher
- Polyalphabetic Cipher
- Vigenere cipher (Autokey system)
- Vernam cipher
- One time pad

1. Caesar cipher:

In cryptography, a Caesar cipher, also known as a Caesar's cipher, the shift cipher, Caesar's code or Caesar shift, is one of the simplest and most widely known encryption techniques.

It is a type of substitution cipher. It is a type of substitution cipher in which each letter in the plaintext is replaced by a letter some fixed number of positions down the alphabet.

For example, with a shift of 3, A would be replaced by D, B would become E, and so on.



The encryption can also be represented using modular arithmetic by first transforming the letters into numbers, according to the scheme,

a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12
n	o	p	q	r	s	t	u	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

- Encryption of a letter by a shift n can be described mathematically as

$$C = E(k, P) = (P + k) \bmod 26$$

- Decryption is performed similarly,

$$P = D(k, C) = (C - k) \bmod 26$$

- Plain text :: meet me after the toga party
- Key :: 3

- Cipher text :: phhw ph diwhu wkh wrjd sduwb
- **Apply Caesar Cipher**
 - Good morning friends welcome to piet KEY = 5
 - Meet me at the usual place at ten rather than eight oclock KEY = 14

Example:

Key	
1	oggv og chvgt vjg vqic rctva
2	nffu nf bgufs uif uphb qbsuz
3	meet me after the toga party
4	ldds ld zesdq sgd snfz ozqsx
5	kecr kc ydrcp rfc rmey nyprw
6	jbbq jb xcqbo qeb qldx mxoqv
7	iaap ia wbpap pda pkcw lwnpu
8	hzzo hz vaozm ocz ojby kmot
9	gyyn gy uznyl nby niau julns
10	fxxm fx tymxk max mhzt itkmr
11	ewwl ew sxlwj lzw lgys hsjlg
12	dvvk dv rkwvi kyv kfyr grikp
13	cuuj cu qvjuh jxu jewq fqhjo
14	btiti bt puitg iwt idvp epgin
15	assh as othsf hvs hcuo dofhm
16	zrrg zr nsgre gur gbtn cnegl
17	yqqf yq mrfqd ftq fasm bmdfk
18	xppe xp lqepc esp ezrl alcej
19	wood wo kpdob dro dyqk zkbdi
20	vnnc vn jocna cqn expj yjach
21	ummb um inbmz bpm bwol xizbg
22	tlla tl hmaly aol avnh whyaf
23	skkz sk glzky znk zumg vgxze
24	rjyy rj fkyjw ymj ytlf ufwyd
25	qiix qi ejxiv xli xake tevxc

- Problem with Caesar Cipher
 - Brute-force attack is possible by simply try all the 25 possible keys.
 - Ex. Cipher text :

phhw ph diwhu wkh wrjd sduwb

- Break cipher text "GCUA VQ DTGCM"

- To makes brute-force cryptanalysis impractical
 - Use large number of keys
 - Language of plaintext is unknown
 - Input file (plaintext) is compressed in some fashion

2. Monoalphabetic cipher

- With only 25 possible keys, the Caesar cipher is far from the security
- If instead of using 26 alphabetic character in sequences, if we do the permutation of 26 character then $26! = 4 \times 10^{26}$ keys are possible and it is difficult to check all those key and this approach is known as monoalphabetic substitution cipher.

A	→	M	N	→	D
B	→	N	O	→	S
C	→	B	P	→	A
D	→	V	Q	→	P
E	→	C	R	→	O
F	→	X	S	→	I
G	→	Z	T	→	U
H	→	L	U	→	Y
I	→	K	V	→	T
J	→	J	W	→	R
K	→	H	X	→	E
L	→	G	Y	→	W
M	→	F	Z	→	Q

■ Plain text ::
meet me after party

■ Cipher text ::
fccu fc mxuco amouw

Attacks on monoalphabetic techniques:

- If cryptanalyst knows the nature of the plaintext (e.g. Noncompressed English text) then analyst can exploit the regularities of the language
- First the relative frequency of the letters can be determined and compared to a standard frequency distribution for English
- If message were long enough, this technique is sufficient, and if short message, we cannot expect exact match.

3. Hill Cypher:

This multi-letter cipher is developed by the mathematician Lester hill in 1929. It takes m successive plaintext letters and substitutes for them m cipher text letters.

- $C = KP \pmod{26}$
- $P = K^{-1}C \pmod{26}$

■ For M=3

■ $C_1 = (k_{11}p_1 + k_{12}p_2 + k_{13}p_3) \pmod{26}$

■ $C_2 = (k_{21}p_1 + k_{22}p_2 + k_{23}p_3) \pmod{26}$

■ $C_3 = (k_{31}p_1 + k_{32}p_2 + k_{33}p_3) \pmod{26}$

$$\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \\ k_{31} & k_{32} & k_{33} \end{pmatrix} \begin{pmatrix} p_1 \\ p_2 \\ p_3 \end{pmatrix} \pmod{26}$$

- Example: encrypt 'meet b' using 2x2 hill cipher with the key $k = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$
- Plain text will be written as ME ET BX
- Letters with there numerical values are as follows

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Numerical values for plain text letters are 12 4, 4 19, 1 23

■ **ENCRYPTION**

$$\begin{bmatrix} c1 \\ c2 \end{bmatrix} = \begin{bmatrix} k11 & k12 \\ k21 & k22 \end{bmatrix} \times \begin{bmatrix} p1 \\ p2 \end{bmatrix} \pmod{26}$$

$$\begin{bmatrix} c1 \\ c2 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix} \times \begin{bmatrix} 12 \\ 4 \end{bmatrix} \pmod{26}$$

- $C1 = (36+4) \pmod{26} = 14 = o$
- $C2 = (60+8) \pmod{26} = 16 = q$
- $C3 = (12+19) \pmod{26} = 5 = f$
- $C4 = (20+38) \pmod{26} = 6 = g$
- $C5 = (3+23) \pmod{26} = 0 = a$
- $C6 = (5+46) \pmod{26} = 25 = z$

■ Encrypted text is : 'oq fg az'

■ **DECRYPTION**

$$\begin{bmatrix} p1 \\ p2 \end{bmatrix} = \begin{bmatrix} k11 & k12 \\ k21 & k22 \end{bmatrix} \times \begin{bmatrix} C1 \\ C2 \end{bmatrix} \pmod{26}$$

$$\begin{bmatrix} p1 \\ p2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -5 & 3 \end{bmatrix} \times \begin{bmatrix} 14 \\ 16 \end{bmatrix} \pmod{26}$$

- $x1 = (28-16) \pmod{26} = 12 = m$
- $x2 = (-70+48) \pmod{26} = 4 = e$
- $x3 = (10-6) \pmod{26} = 4 = e$
- $x4 = (-25+18) \pmod{26} = 19 = t$
- $x5 = (0-25) \pmod{26} = 1 = b$
- $x6 = (0+75) \pmod{26} = 23 = x$

■ Decrypted text is : 'me et bx'

Inverse of matrix

- Encrypt a message "CIPHER" using 3x3 hill cipher with key=[{2,1,1}, {1,1,2}, {1,0,-2}]

$$K = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 0 & -2 \end{bmatrix}$$

$$K^{-1} = \begin{bmatrix} 2 & -2 & -1 \\ -4 & 5 & 3 \\ 1 & -1 & -1 \end{bmatrix}$$

- Plain text: CIP HER
- Cipher text: BOY JTZ

4. Playfair Cipher

- The best-known multiple-letter cipher is the Playfair
- Playfair cipher is based on use of 5 by 5 matrix letters constructed using a keyword

- To generate the key table, one would first fill in the spaces in the table with the letters of the keyword (dropping any duplicate letters), then fill the remaining spaces with the rest of the letters of the alphabet in order from left to right and from top to bottom
- The letters I and J count as one letter
- Plaintext is encrypted two letters at a time

■ Key phrase: KEYWORD

K	E	Y	W	O
R	D			

K	E	Y	W	O
R	D	A	B	C
F	G	H	IJ	L
M	N	P	Q	S
T	U	V	X	Z

Rules:

- To encrypt a message, one would break the message into digraphs (groups of 2 letters) such that,
 - Ex , "Hello World" becomes "HE LL OW OR LD"
- If both letters are the same in pair (or only one letter is left), add an "X" after the first letter. Some variants of Playfair use "Q" instead of "X",
 - Ex. "HE LL OW OR LD" would be treated as HE LX LO WO RL DX
- If the two plaintext letters appear on the same row of your table, replace them with the letters to their immediate right respectively (with the first element of the row circularly following the last.)
 - Ex. BC is encrypted as CR
- If the two plaintext letters appear on the same column of your table, replace them with the letters immediately below respectively (with the first element of the column circularly following the last.)
 - Ex. KT is encrypted as RK
- Otherwise, each plaintext letter in a pair is replaced by the letter that lies in its own row and the column occupied by the other plaintext letter.
 - Ex DQ is encrypted as BN
 - Ex AS is encrypted as CP

Example:

- Plain text: why don't you

- WH YD ON TY OU

K	E	Y	W	O
R	D	A	B	C
F	G	H	IJ	L
M	N	P	Q	S
T	U	V	X	Z

- Cipher text: YI EA ES VK EZ

Example:

- Plain text: IMPOSSIBLE
- IM PO SS IB LE
- Regrouping: IM PO SX SI BL EX

K	E	Y	W	O
R	D	A	B	C
F	G	H	IJ	L
M	N	P	Q	S
T	U	V	X	Z

- Cipher text: FQ SY QZ QL CI WU
- Plain text : IM PO SX SI BL EX

Example(6X6 matrix):

- Key: KEY WORD

K	E	Y	W	O	R
D	A	B	C	F	G
H	I	J	L	M	N
P	Q	S	T	U	V
X	Z	0	1	2	3
4	5	6	7	8	9

- Plain text: BECSE ROOM NO 325
- BE CS ER OO MN O3 25
- Regrouping: BE CS ER OX OM NO 32 5X
- Cipher text: AY BT YK K2 FU MR X3 4Z

- Great advance over mono-alphabetic cipher as 26 letters, there are $26 * 26 = 676$ diagrams so

identification of individual diagrams is more difficult

- Relative frequencies of individual letters exhibits much greater range so frequency analysis much more difficult
- For these reason it was for long time considered unbreakable
- It was used by British Army in world war I and by U.S Army in world war II.
- Still the Playfair Cipher is relatively easy to break, because it leaves much of the structure of the plaintext language intact. A few hundred letters of cipher text are generally sufficient.

5. Polyalphabetic Cipher

VIGENERE CIPHER

- Another way to improve on the simple monoalphabetic technique is to use different monoalphabetic substitutions as one proceeds through the plaintext message known as **polyalphabetic cipher**
- The best and simplest algorithm is Vigenere cipher
- Each cipher is denoted by a key letter, which is the cipher text letter that substitutes for the plaintext letter
- To encrypt, a table of alphabets can be used, termed a Vigenere tableau, *Vigenère square*, or *Vigenère table*
- Each of the 26 ciphers is laid out horizontally, with the key letter for each cipher to its left. A normal alphabet for the plaintext runs across the top.
- **Encryption Process** : Given a key letter k and plaintext letter p, the cipher text letter is at the intersection of the row labeled k and the column labeled p.
- **Decryption Process** : Given a key letter identifies the row. The position of the cipher text letter in that row determines column and the plaintext letter is at top of that column
- To encrypt: the key is as long as the message and it is a repeating keyword

- Plain text: ATTACKATDAWN
- Key: LEMON

A	T	T	A	C	K	A	T	D	A	W	N
L	E	M	O	N	L	E	M	O	N	L	E

Plaintext

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Cipher text :LXFOPVEFRNHR

- Plaintext: information security
- Key: confidential
- Encryption: KBSTZPEGBWNDGQHWQWC
- Plaintext: crypto is for cryptography
- Key: abcdef
- Encryption: CSASXTITHRVHRZRWSLRBRKC

VERNAM CIPHER

- The ultimate defense against such cryptanalysis is to choose keyword that is as long as the plaintext and has no statistical relationship to it.
- Gilbert vernam in 1918 devised a system that works on binary data rather than letters.

Encryption : $c_i = m_i \oplus k_i$ $i = 1, 2, 3, \dots$

m_i : plain-text bits.

k_i : key (key-stream) bits

c_i : cipher-text bits.

Decryption : $m_i = c_i \oplus k_i$ $i = 1, 2, 3, \dots$

- Essence is creation of the key

SENDING

```
-----
message: 0 0 1 0 1 1 0 1 0 1 1 1 ...
pad:     1 0 0 1 1 1 0 0 1 0 1 1 ...
XOR
cipher:  1 0 1 1 0 0 0 1 1 1 0 0 ...
```

RECEIVING

```
-----
cipher:  1 0 1 1 0 0 0 1 1 1 0 0 ...
pad:     1 0 0 1 1 1 0 0 1 0 1 1 ...
XOR
message: 0 0 1 0 1 1 0 1 0 1 1 1 ...
```

6. OTP (One Time Pad):

- It is proposed to improvement in vernam cipher.
- Use of random Key that is as long as the message and not repeated.
- Key is used to encrypt & Decrypt single message and then it is discarded.
- Each new message requires new key of same length.
- It produces random output that bears no statistical relationship to the plaintext so there is no way to break the code.
- So OTP is unbreakable.

Ex : Suppose we are using vigenere scheme with 27 character in which 27th character is blank space the table is

	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	
	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	
a	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	
b	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	
c	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	
d	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	
e	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	
f	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	
g	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	
h	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	
i	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	
j	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	
k	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	
l	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	
m	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	
n	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	
o	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	
p	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	
q	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	
r	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	
s	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	
t	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	
u	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	
v	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	
w	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	
x	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	
y	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	
z	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	
	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	

- Consider Cipher text:
 - ANKYODKYUREPFJBYOJDSPLREYIUNOFDOIUERFPLUYTS
- Key:
 - pxlmvmsydofoyrvzwc tlnbncvqgdupahfzzlmnyih
- Plaintext:
 - mr mustard with the candlestick in the hall
- Key :: pftgpmiydgaxgoufhklmhsqdgogtewbqfgyovuhwt
- Plaintext:
 - miss scarlet with the knife in the library
- No way to decide which key is correct and therefore which plaintext is correct
- Two fundamental difficulties:

- 1) Practical problem of making large quantities of random keys.
 - 2) Even more daunting is the problem of key distribution and protection
- For every message to be sent a key of equal length is needed by both sender and receiver

References:

1. Cryptography and Network Security (Textbook) William Stallings; Pearson Education
2. Cryptography & Network Security Behrouz A. Forouzan; Tata McGraw-Hill
3. Information Security Principles and Practice Deven Shah, Wiley-India
4. Information Security Principles and Practice Mark Stamp; Wiley India edition
5. Information systems security Nina Godbole; Wiley Publications, 200

2.4 Transposition Techniques

- Performs some sort of permutation on the plaintext letters
- Some of the Transposition Techniques are
 - Rail Fence Technique
 - Simple transposition-permutation
 - Reverse transposition cipher
 - Odd-even transposition technique

Rail fence techniques

- Plaintext is written down as a sequence of diagonals and then read off as a sequence of row.
- Plain text: meet at five pm behind p lab.
- Written as rail fence of depth 2:

M		e		a		f		v		p		b		h		n		p		a	
	e		t		t		i		e		m		e		i		d		l		b

- Encrypted as: meafvpbhnpaettiemeidlb
- With rail fence of depth 3:
- Plaintext : 'WE ARE DISCOVERED. FLEE AT ONCE',

W	.	.	.	E	.	.	.	C	.	.	.	R	.	.	.	L	.	.	.	T	.	.	.	E
.	E	.	R	.	D	.	S	.	O	.	E	.	E	.	F	.	E	.	A	.	O	.	C	.
.	.	A	.	.	I	.	.	.	V	.	.	.	D	.	.	.	E	.	.	.	N	.	.	.

- Encrypted as: WECRLTEERDSOEFEAOCAIVDEN
- A more complex scheme is to write a message in a rectangle (square matrix), row-by-row and read it off, column by column.
- Order of the columns then becomes the key.
- Pain text: meet at five pm behind p lab.

m	e	e	t	a
t	f	i	v	e
p	m	b	e	h
i	n	d	p	l
a	b	x	x	x

- Encrypted text: mtpiaefmnbeibdxtvepxaehlx

- Alternatively, a key can also be defined to permute the order of the column, e.g. encryption key (41523) defines write in a row-by-row, read it off: 4th column first, 1st column second, 5th column third, followed by 2nd and 3rd column, and prepare encrypted text.

References:

1. Cryptography and Network Security (Textbook) William Stallings; Pearson Education
2. Cryptography & Network Security Behrouz A. Forouzan; Tata McGraw-Hill
3. Information Security Principles and Practice Deven Shah, Wiley-India
4. Information Security Principles and Practice Mark Stamp; Wiley India edition
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2.5 Steganography

- Steganography is the art and science of writing hidden messages in such a way that no one apart from the sender & intended recipient knows of the existence of the message.
- The word steganography is of Greek origin words steganos (στεγανός), meaning "covered or protected", and graphei (γραφή) meaning "writing".
- Cryptography relies on transformation algorithms using key to scramble a message.
- Steganography takes one piece of information and hides it within another.
- Both are used to protect information but steganography is concerned with hiding information thereby making it unseen while cryptography is concerned with encrypting information thereby making it unreadable.
- Some Examples:
 - **Character marking:** where selected letters of text are overwritten with a special pencil. The marks are ordinarily not visible unless the paper is held at an angle to bright light.
 - **Use of invisible ink.** Visible when some heat or chemical applied
 - **Pin punctures.** On selected letters visible in front of light
 - **Typewriter correction ribbon** visible under strong light
- Types of Steganography:
 - **Audio steganography**
 - **Video steganography**
 - **Textual steganography**
 - **Image steganography**

References:

1. Cryptography and Network Security (Textbook) William Stallings; Pearson Education
2. Cryptography & Network Security Behrouz A. Forouzan; Tata McGraw-Hill
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